



Inspired DATA+AI Applications

AI for Image Analytics

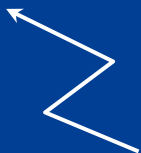
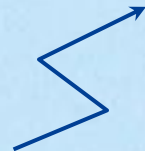
Case Study and Research



AI Reality Lab

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Assistant Professor @GSAS NIDA
AI Reality Lab



Associate Professor @GSCM NIDA
AI Reality Lab



Dr.Chutisant Kerdvibulvech

A Landscape

The big picture of our researches
and case studies



DATA+Artificial Intelligence

"..an area of computer science that emphasizes the creation of intelligent machines that work and react like humans.." AI Definition from Gunner Technology



Thai PBS: https://www.youtube.com/watch?v=GDrLyzd_Tsc



Artificial Intelligence



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- 
- **Computer Vision (CV): Image Analytics**
 - **Natural Language Processing (NLP): Text Analytics**
 - Others: Speech Recognition, etc.
- 
- 
- 

Our Researches in AI Reality Lab



Our Researches in AI Reality Lab

1. HUMAN



3. SOCIAL MEDIA

2. OBJECT



4. GAN



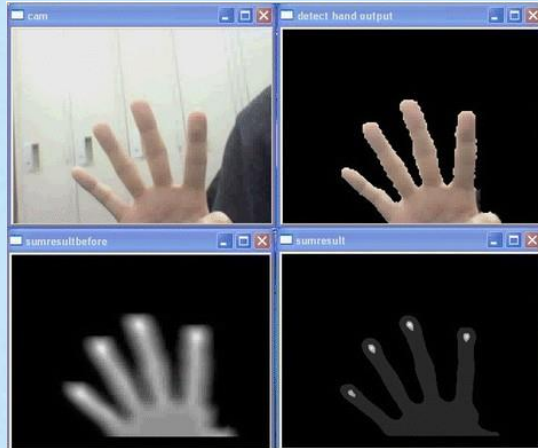


Our Researches: Image Analytics

\$

1. HUMAN

1.1 Human Hand



EURASIP Journal on Embedded Systems, 2014

Particle Filter



Lecture Notes in Computer Science, 2020

Office Syndrome (Hand Therapy)



Journal of KMUTNB, 2021

Liveness Detection





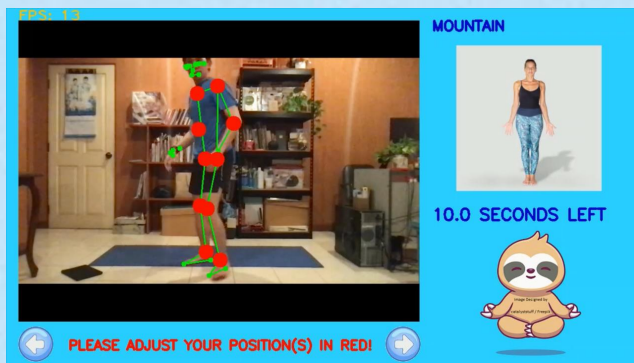
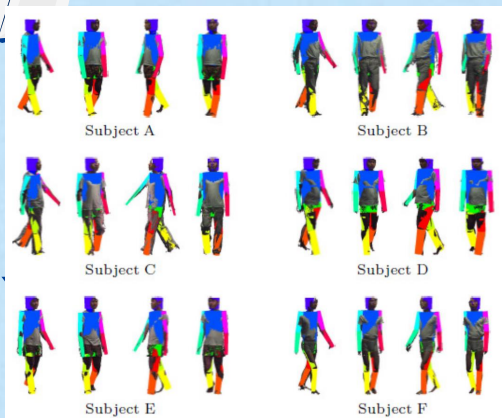
Our Researches: Image Analytics



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1. HUMAN

1.2 Human Full Body



Personal AI Yoga Trainer, 2021



Food Delivery Analytics, 2021

Structural and Syntactic Pattern Recognition, 2014
Human Shape Analysis



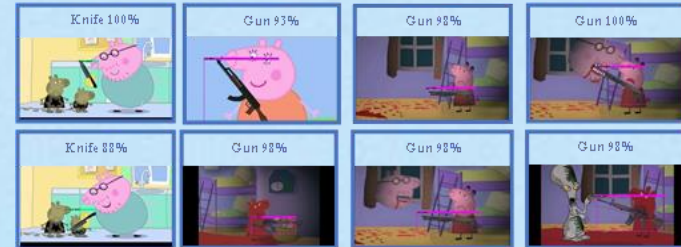
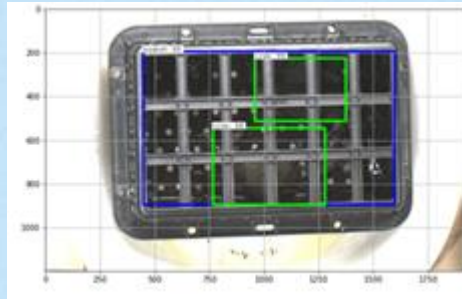
Our Researches: Image Analytics

2. OBJECT

2.1 Violence detection



2.2 Mud Crab



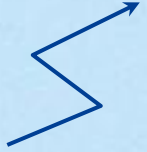
Submitted to Journal, 2021
Cartoon for Kids

Submitted to Journal, 2021

Soft-shell Mud Crab in a Basket Using Deep Learning



Our Researches: Image Analytics

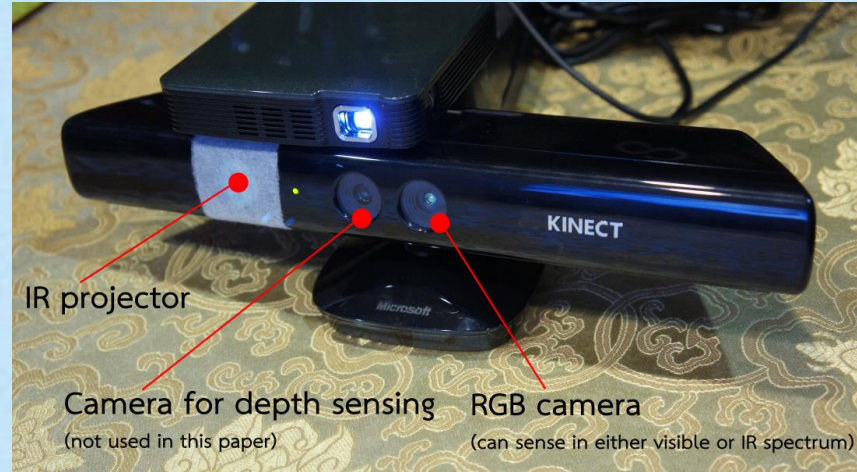


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2. OBJECT



2.3 Augmented Reality



Pacific Rim Conference on Multimedia (PCM), 2015

**Estimate Infrared Vision of Printed Color Using
Bayesian Classifier and K-Nearest Neighbor Regression**





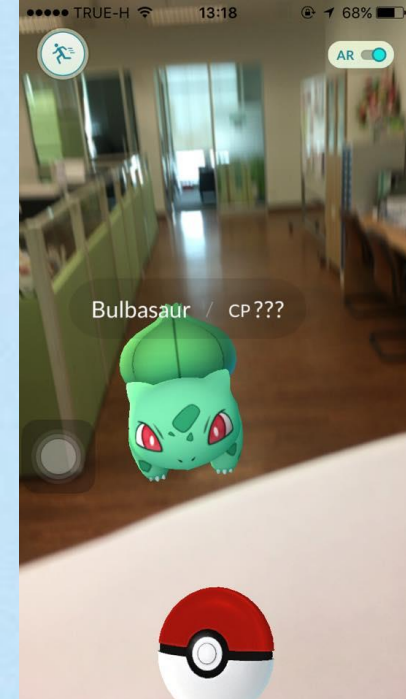
Our Researches: Image Analytics

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IEEE Southwest Symposium on Image Analysis & Interpretation (SSIAI), 2016

Our AR Car Game



By Niantic, 2016

Their Pokémon GO



Pokémon GO Revenue



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The game was awarded five Guinness World Records in August 2016

- 
- 
- Most revenue grossed by a mobile game in its first month (US\$206.5 million)
 - Most downloaded mobile game in its first month (130 million downloads)
 - Most international charts topped simultaneously for a mobile game in its first month (top game in 70 different countries)
 - Most international charts topped simultaneously for a mobile game in its first month (top grossing in 55 countries simultaneously)
 - Fastest time to gross \$100 million by a mobile game (reached in 20 days on July 26)
 - The game had over 147 million monthly active users by May 2018, over a billion global downloads by early 2019, **and grossed more than \$6 billion in revenue as of 2020.**
- 

“COVID-19”



- The game generated **a revenue of \$641.6 million in the first half of 2021.**



Metaverse and Its Revenue



Meta Revenue:
US\$85.97 billion



Facebook Reality
Lab, 2021



CNN, October 2021

Metaverse and Its Revenue



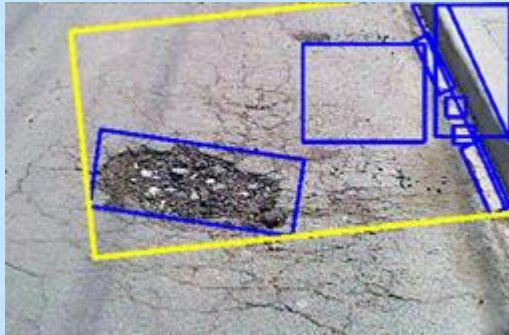
Usually use AI for Image Analytics



Our Researches: Image Analytics

2. OBJECT

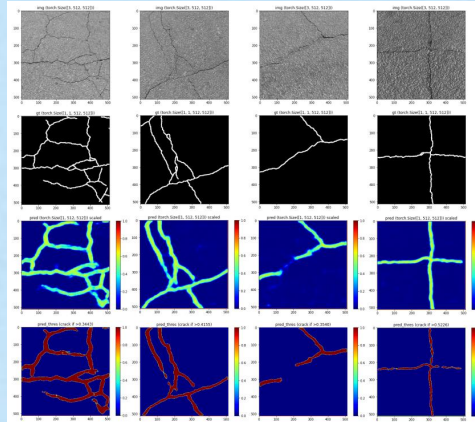
2.4 Segmentation



Advances in Multimedia, 2018

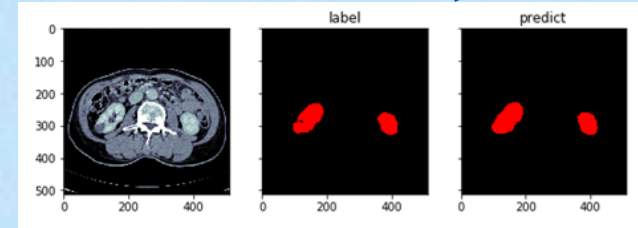
Automatic Road Distress Visual Inspection System

\$



Lecture Notes in Computer Science (LNCS), 2021

Deep Learning-based Using Modified U-Net for Road



Submitted to Journal, 2021

Automatic Kidney Segmentation
based on CT Images





Our Researches: Image Analytics



3. SOCIAL MEDIA

3.1 Caption Recommendation

3.2 Hashtag Analyzer



4. GAN (Generative Adversarial Network)

4.1 Generate a new shoe pattern

4.2 Privacy-preserving Face ID



```
"Gender": {  
  "Confidence": 99.72714233398438,  
  "Value": "Male"  
}
```



```
"Gender": {  
  "Confidence": 99.9794921875,  
  "Value": "Female"  
}
```



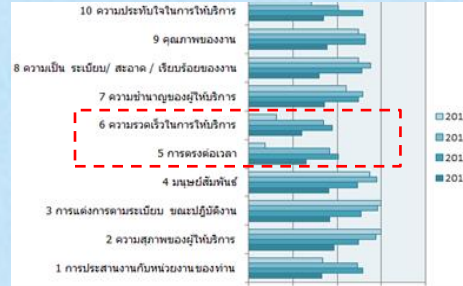
Submitted to Journal, 2021

Privacy-preserving Face ID for Face Recognition



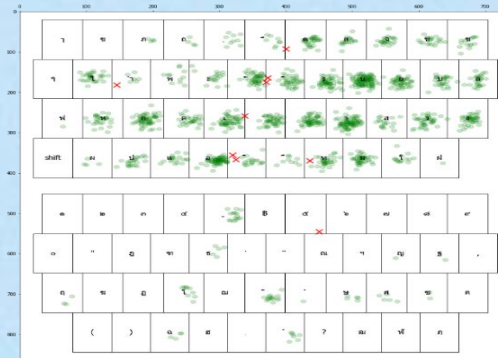
Our Researches: NLP

1. IT Ticket Classification



KMUTT R&D Journal, 2021
**Using Machine Learning
to Improve Ticket
Classification**

2. Hashtag Analyzer



Journal of KMUTNB, 2021
**Reduce Typos in Soft
Keyboard**

3. Depression Detection System (AI Sentiment Analysis)



B Details

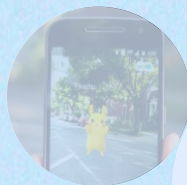
Walking through each work
one-by-one

HUMAN

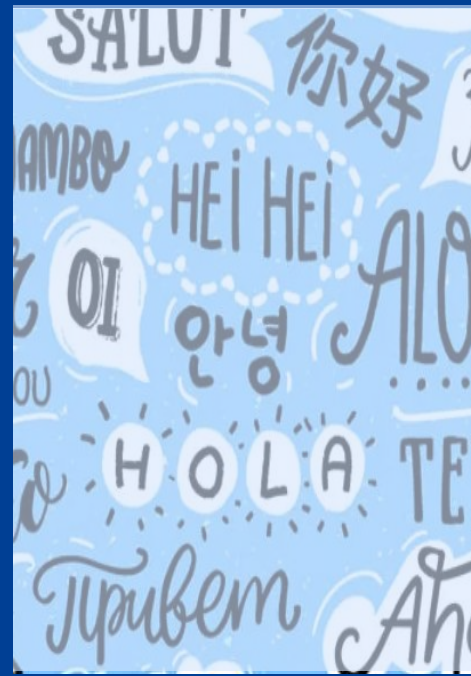


SOCIAL
MEDIA

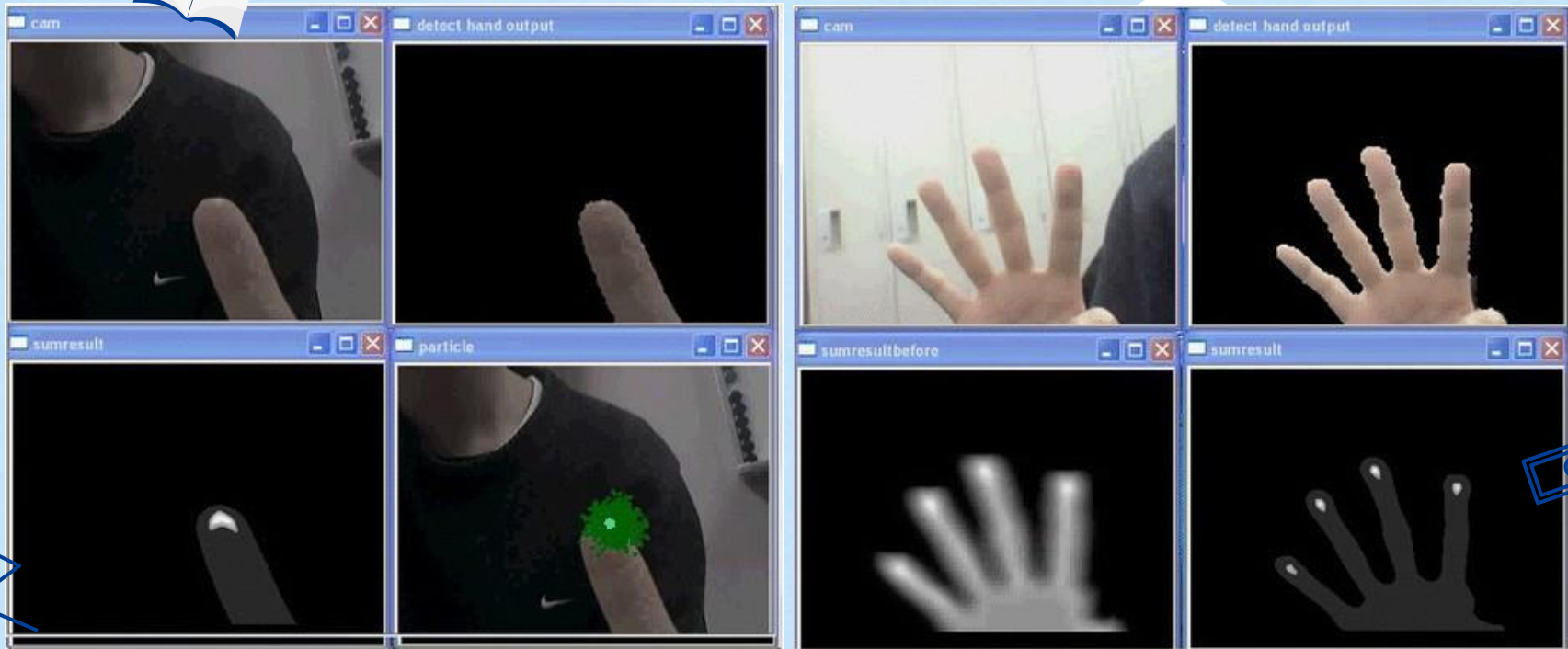
OBJECT



GAN



Human: Hand



Human: Hand



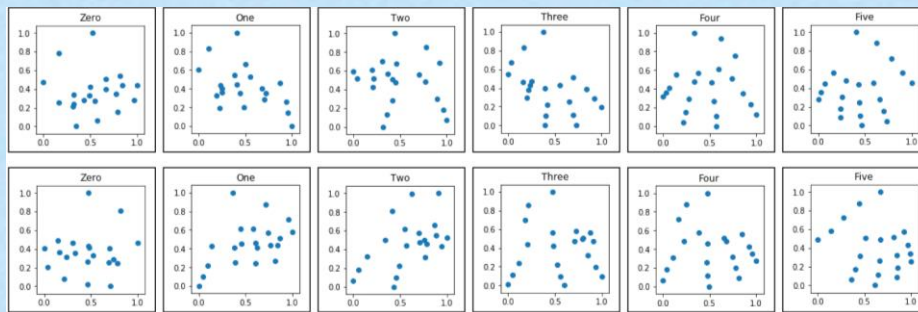
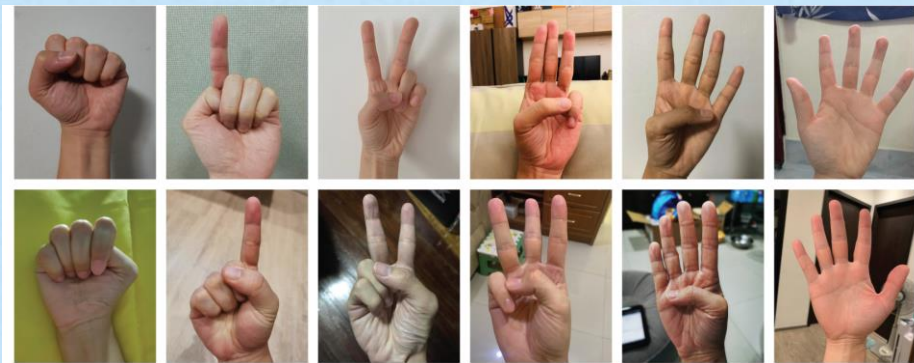
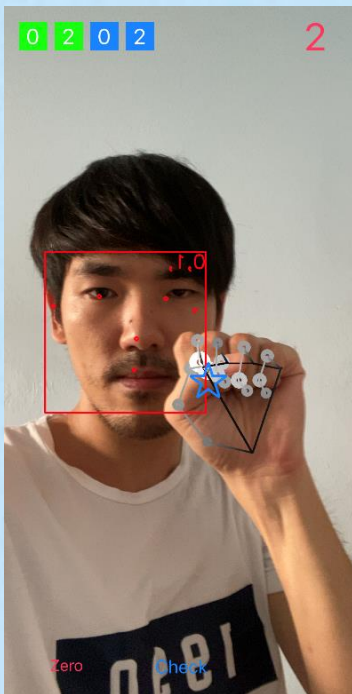
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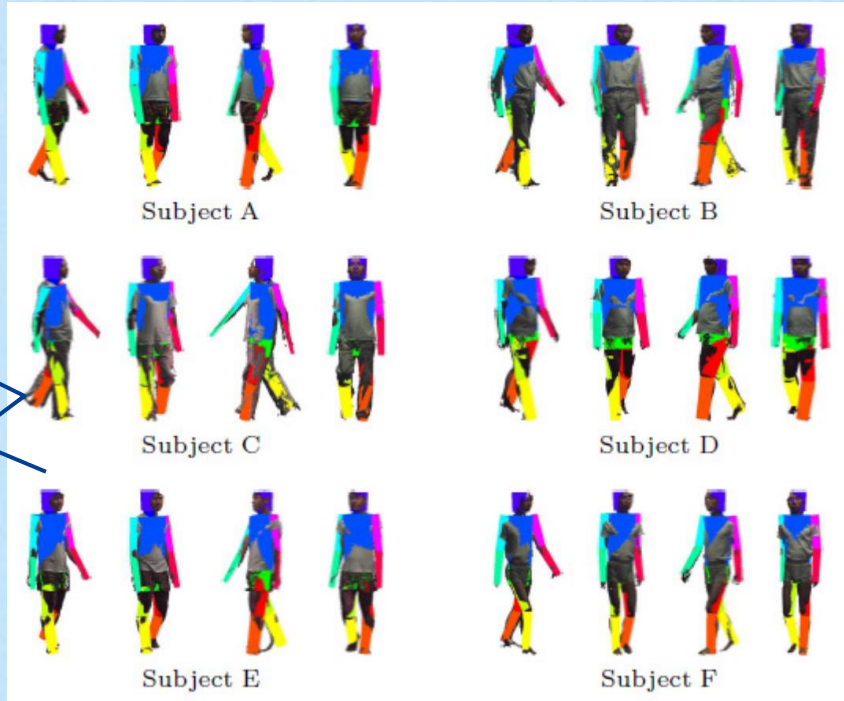
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Human: Hand



Human: Full Body



- Intrinsic and extrinsic parameters

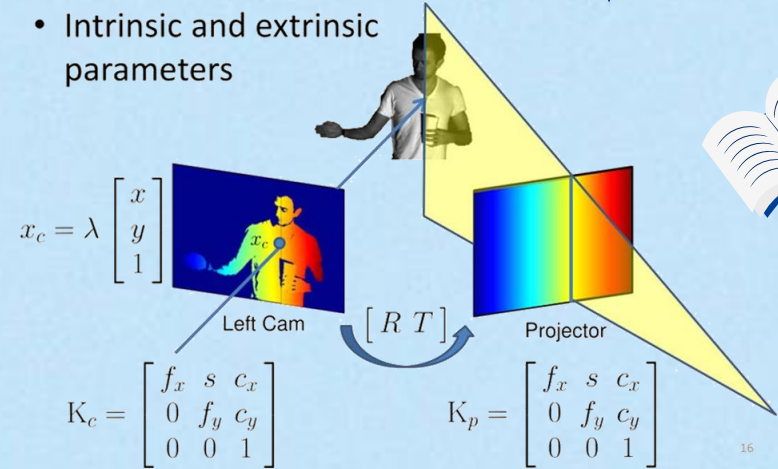
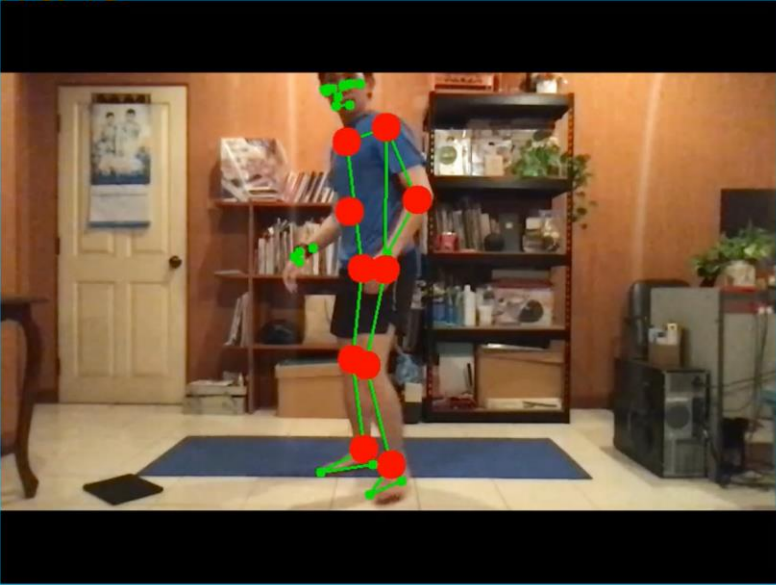


Image credit: <https://www.slideserve.com/whitby/introduction-to-structured-light-sl-systems-and-sl-based-phase-unwrapping-powerpoint-ppt-presentation>

Human: Full Body

FPS: 13



MOUNTAIN



10.0 SECONDS LEFT



PLEASE ADJUST YOUR POSITION(S) IN RED!

Human: Full Body

\$



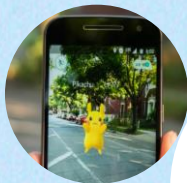
\$

HUMAN

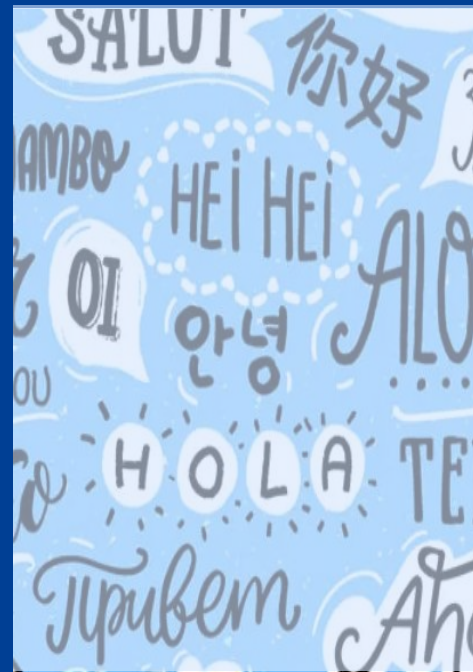


SOCIAL
MEDIA

OBJECT

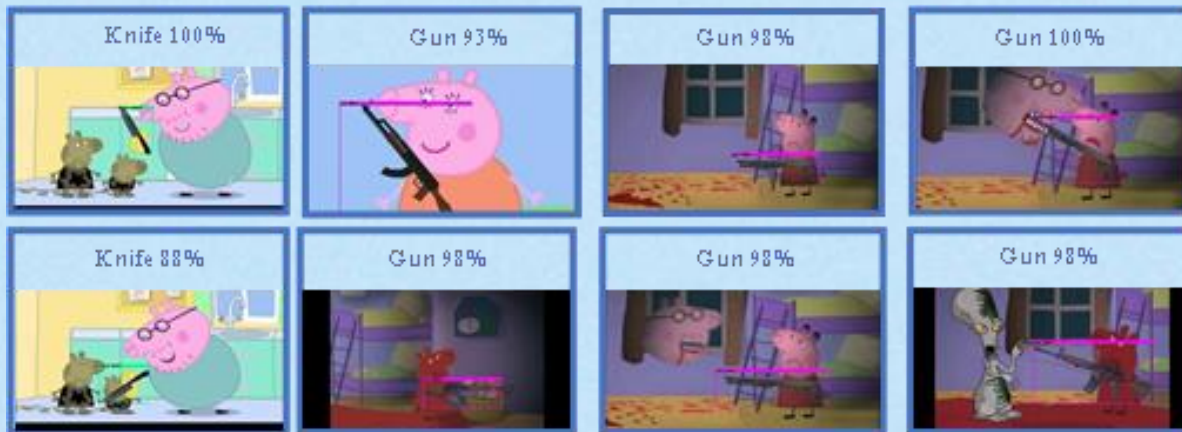
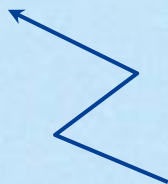


GAN





Object: Violence Detection

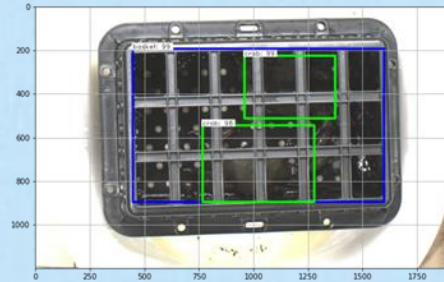
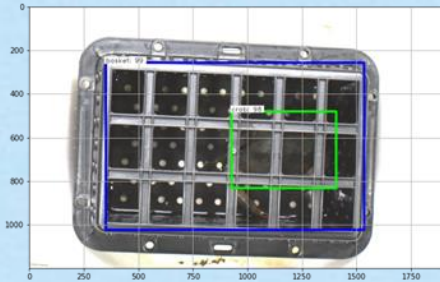
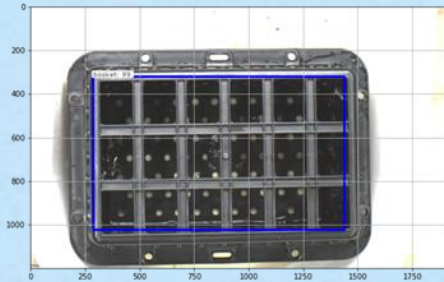
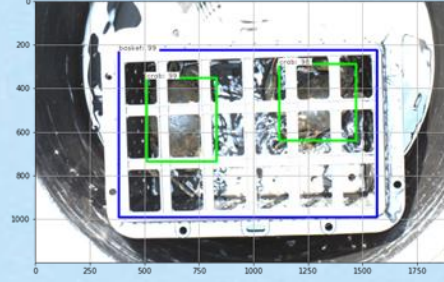
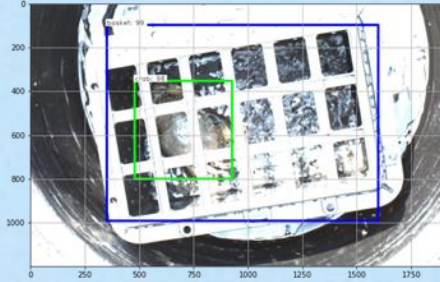
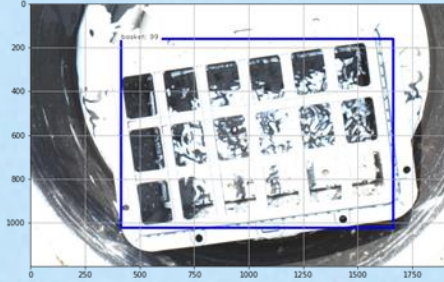


\$



Object: Molted Crab

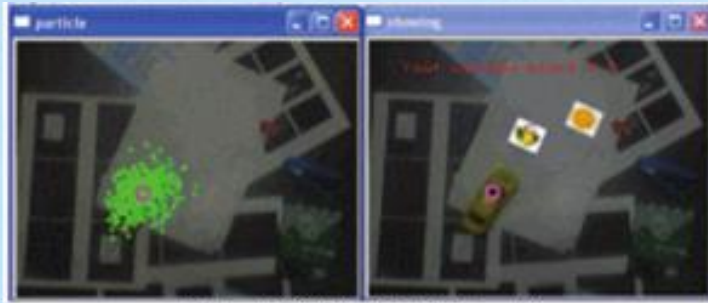
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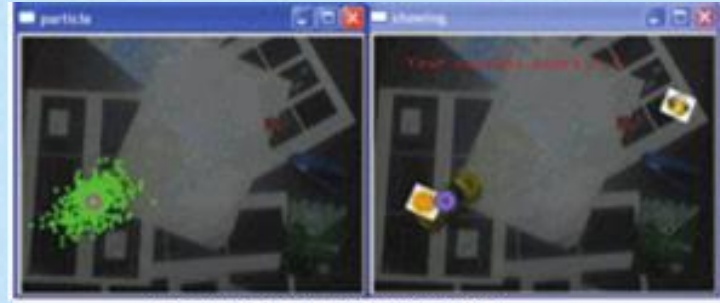


\$

Object: Augmented Reality



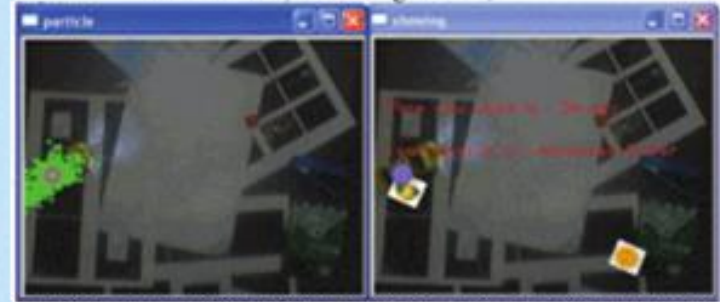
Frame 10 (starting the game)



Frame 28 (collecting a coin)



Frame 15 (collecting a mushroom)



While the game finishes, it evaluates the skill level of a user

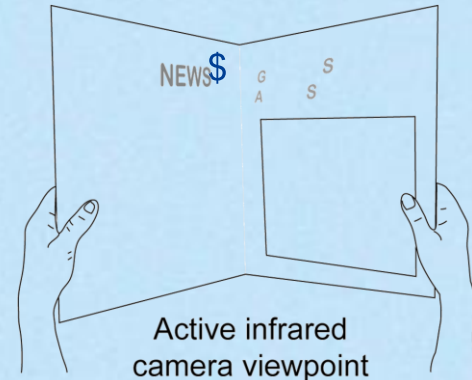
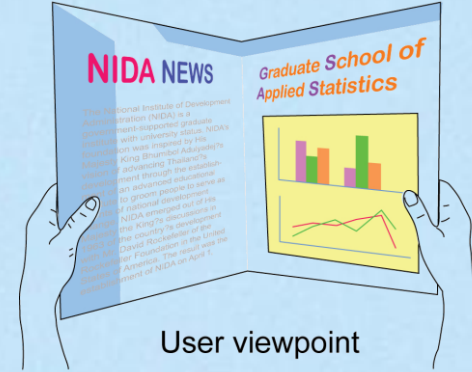
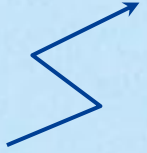




\$

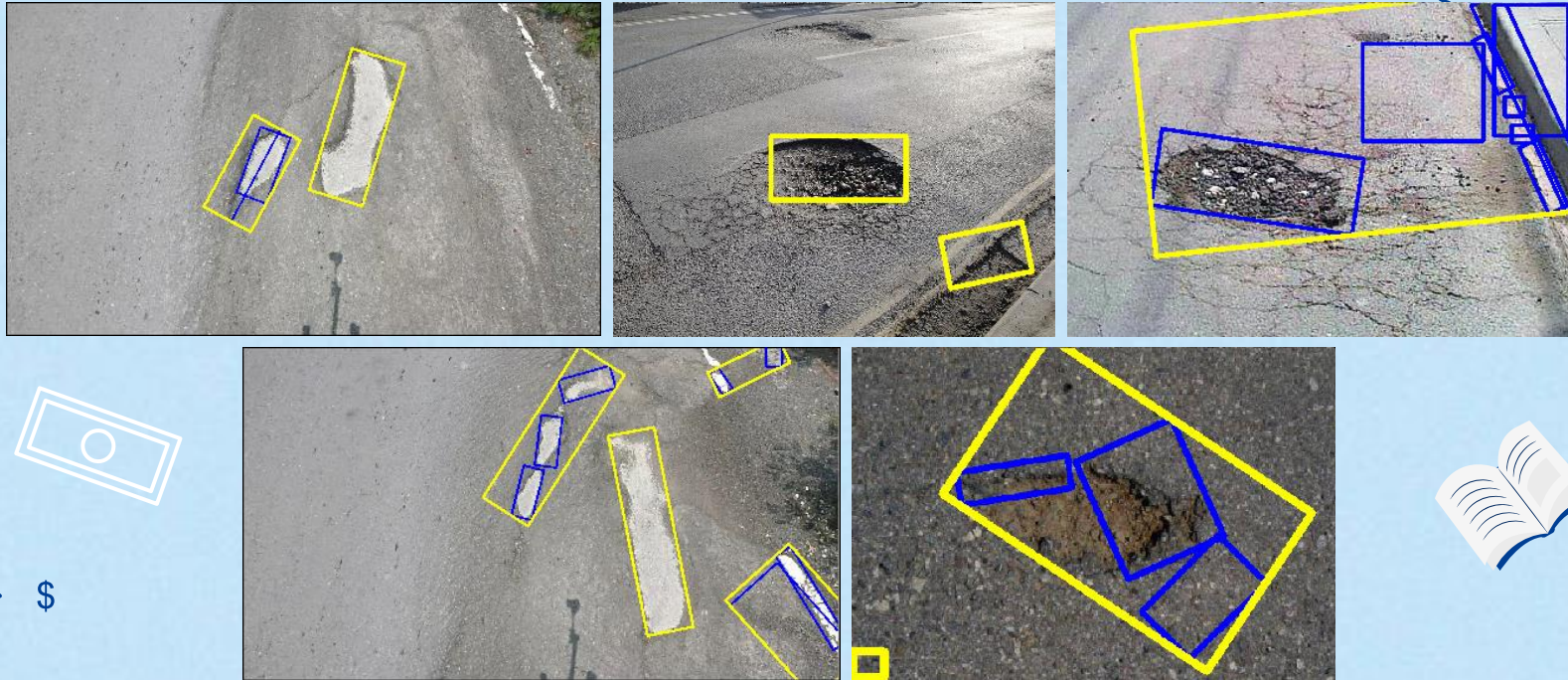


Object: Augmented Reality

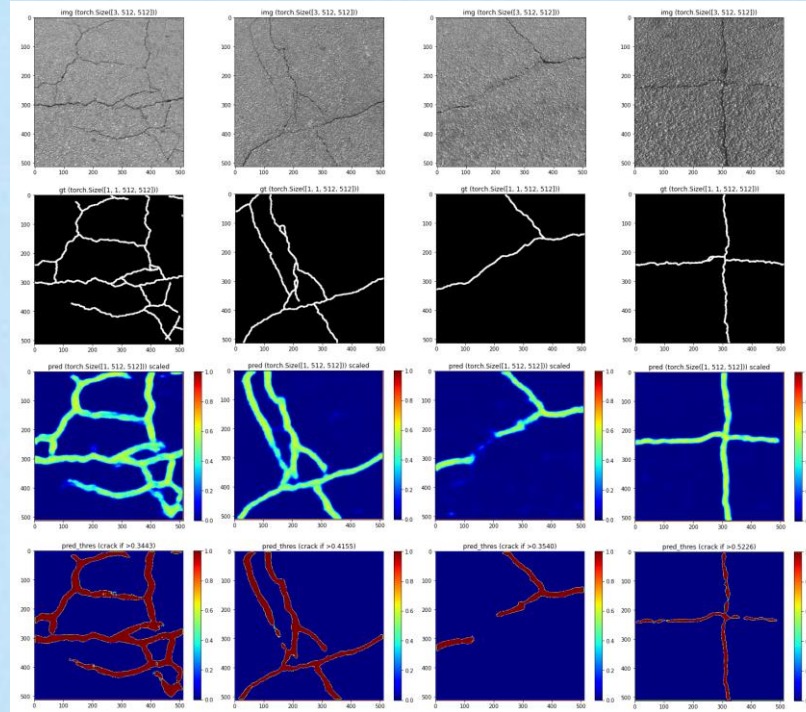


- T. Siriborvornratanakul (2014). **Multimodal Interface Towards Smartphones: The Use of Pico Projector, Passive RGB Imaging and Active Infrared Imaging.** In *International Conference on Articulated Motion and Deformable Objects (AMDO)*.
- T. Siriborvornratanakul (2015). **Color and Active Infrared Vision: Estimate Infrared Vision of Printed Color Using Bayesian Classifier and K-Nearest Neighbor Regression.** In *Pacific Rim Conference on Multimedia (PCM)*.

Object: Segmentation



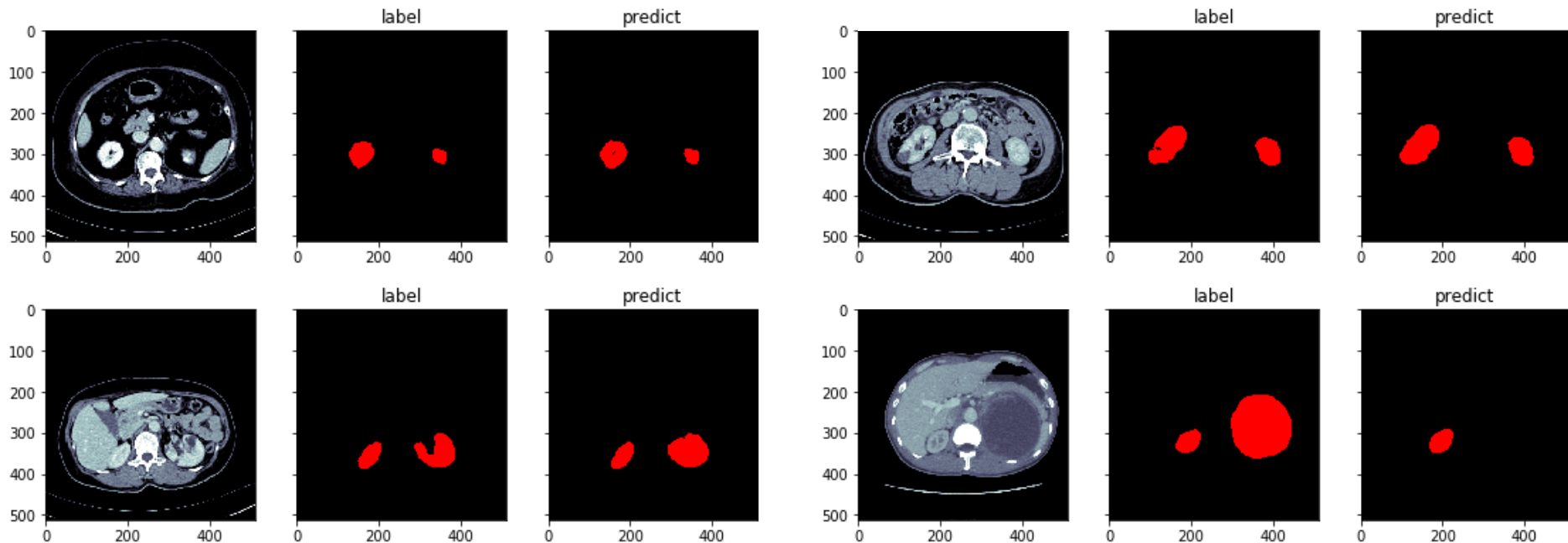
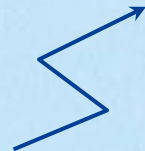
Object: Segmentation





Object: Segmentation

\$



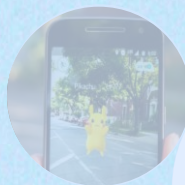
P. Kittipongdaja and T. Siriborvornratanakul. **Automatic Kidney Segmentation Using 2.5D ResUNet and 2.5D DenseUNet for Malignant Potential Analysis in Complex Renal Cyst based on CT Images.** (under review)

HUMAN

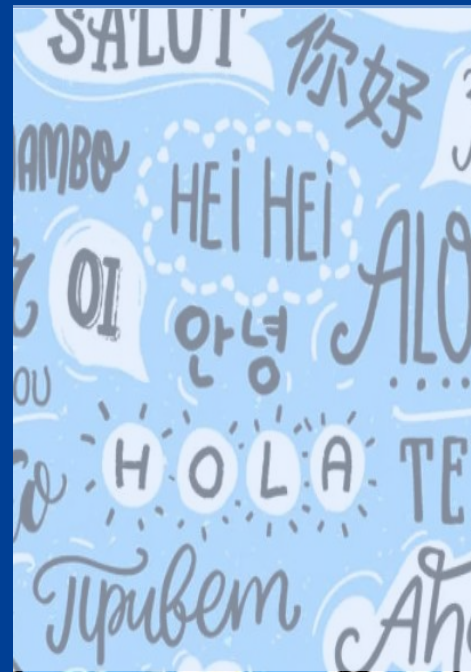


SOCIAL
MEDIA

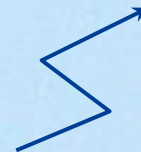
OBJECT



GAN



Social Media: Caption Recommendation



\$



\$



ความสุขของชานคือการมองน้องหมา : 110.64
บ่มน่ารักโหมครีบ สุนัขขี้อ้อน : 103.25
เวลาเจอเปิดมันร้องก๊ากก้าบ !!แต่ร้องรักค้าบเวลาเจอคุณ : 95.13
แค่คุณรักหมา ก็ได้ใจเราไปครึ่งหนึ่งละนะ : 88.86
ยิ่งฉันอยู่กับคนมากเท่าไร ฉันก็ยิ่งรักสุนัขมากขึ้นเท่านั้น : 74.98

\$

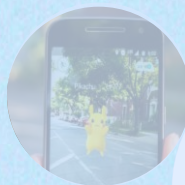


HUMAN

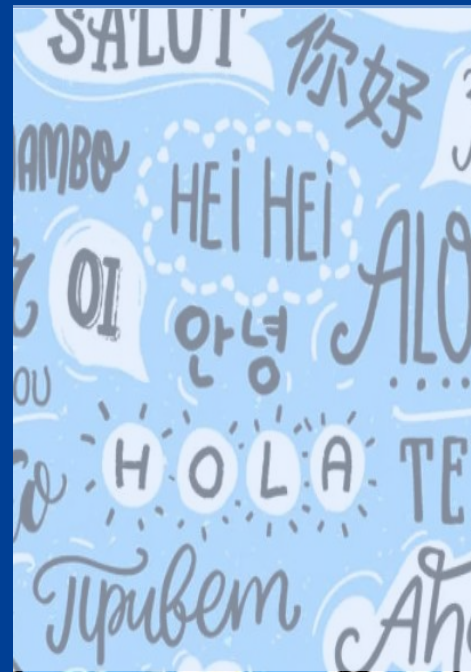


SOCIAL
MEDIA

OBJECT



GAN

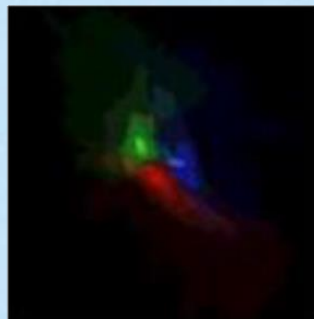




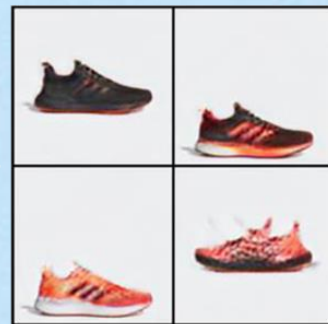
GAN: Generate a new shoe pattern



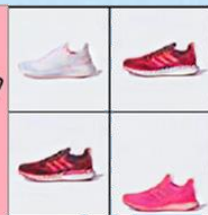
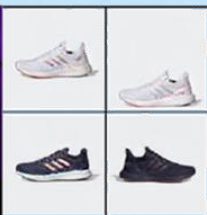
Target image



Histograms
(Hidden process in HistoGANs)



GAN-generated images based on
specified histogram feature



GAN: Privacy-preserving Face ID



```
"Gender": {  
  "Confidence": 99.72714233398438,  
  "Value": "Male"  
}
```

```
"Gender": {  
  "Confidence": 99.9794921875,  
  "Value": "Female"  
}
```

\$



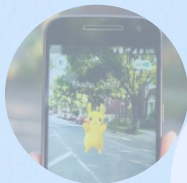
\$

HUMAN



SOCIAL
MEDIA

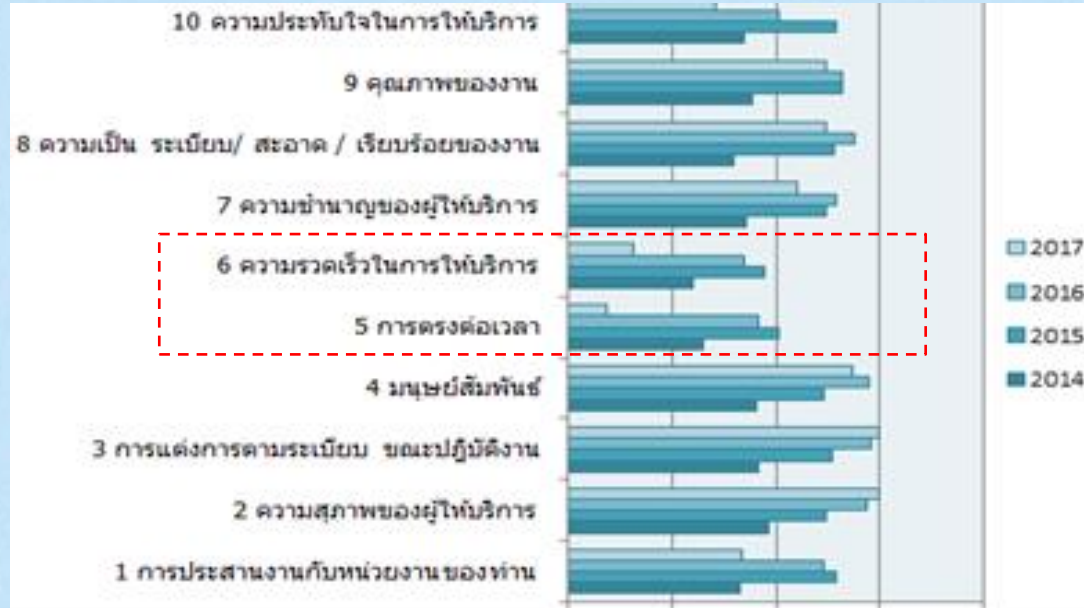
OBJECT



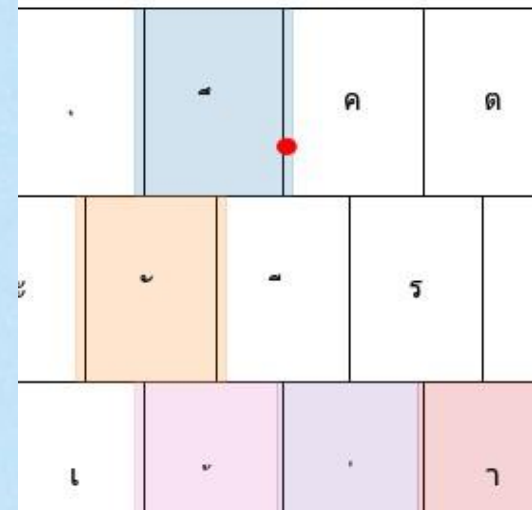
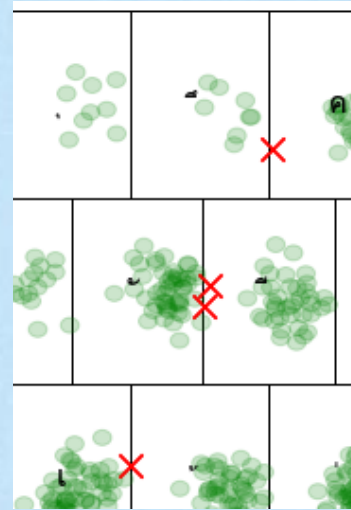
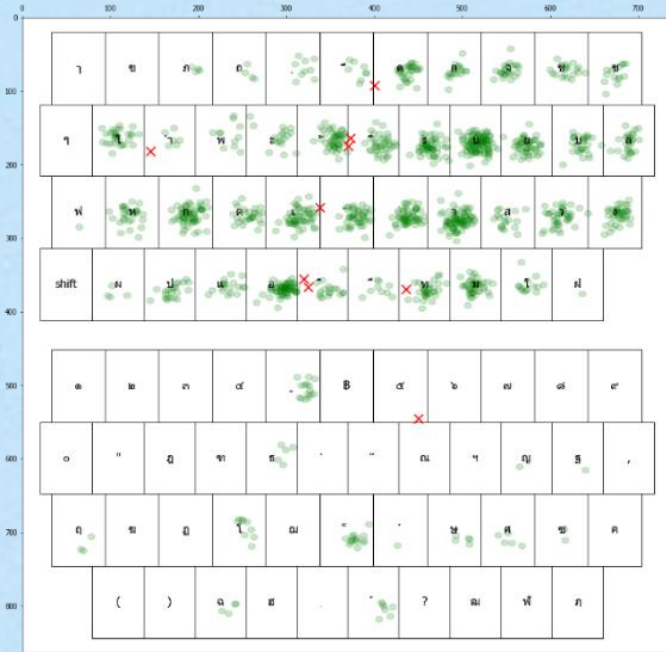
GAN



NLP: IT Ticket Classification



NLP: Key-target Resizing





NLP: Depression Detection System

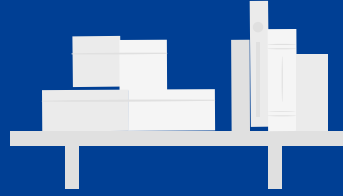
\$

- AI Sentiment Analysis
- Social Media Analysis: Facebook and Instagram
- Post, Share, Emotion, #, Story, Check-in, Live
- In Process





\$



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C

Conclusion

Last but not least for today's presentation



2012



2018



MEETING ROOMS

3 Meetings per day

37%

Utilization

23%

IS consumed per day

72% Sitting time daily

DESKS

78%

Utilization

11%

27 cups brewed



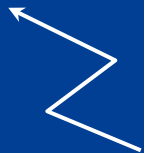
Q&A



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Live
ONLINE TRAINING

DATA + AI
POWERED COMMUNICATION 2022



Day 2

UNDERSTANDING DATA+AI

เจาะลึกเทคโนโลยีปัญญาประดิษฐ์
จากผู้เชี่ยวชาญปัญญาประดิษฐ์ตัวจริงระดับประเทศ

11 พฤศจิกายน 2564 | 09.00 - 16.00 น.

Impact of DATA+AI on the Future
of Work and Society



ดร.กอบกฤตย์ วรียะยุทธกร

นายกสมาคมผู้ประกอบการ
ปัญญาประดิษฐ์ประเทศไทย
(AI-EAT)



รศ.ดร.อัศวิน เนตรโพธิ์แก้ว

ผู้ช่วยอธิการบดีฝ่ายสื่อสารองค์กร
และกิจการนักศึกษา และอาจารย์ประจำ
คณะนิติศาสตร์และบริหารการศึกษาร NIDA

DATA+AI Readiness
for Organization



คุณจรัส งามวิโรจน์เจริญ

Director (CEO Office)
CJ Express Group

Fundamental DATA+AI Concepts:
Workshop and Tools for Deep Learning



รศ.ดร.ชุตีสัตย์ เกิดวิบูลย์โส

คณะนิติศาสตร์และบริหารการศึกษาร
สถาบันบัณฑิตพัฒนบริหารศาสตร์



ผศ.ดร.ชุตีสัตย์ ศิริวรรณกุล

คณะสังคมวิทยาและมานุษยวิทยา
สถาบันบัณฑิตพัฒนบริหารศาสตร์



ดูรายละเอียด
ลงทะเบียน
สแกนเลย



จัดโดย นิตะ@นิต้า CIC

ได้รับการสนับสนุนจาก กองทุนส่งเสริมวิทยาศาสตร์
วิจัยและนวัตกรรม (ววน.)



Nitade@NIDA
Live Talk/Teach

“สร้างปัญญาประดิษฐ์ (AI) อย่างไรไม่ให้ผิดกฎหมาย”

วันเสาร์ที่ 20 พฤศจิกายน 2564 เวลา 16.00 - 17.00 น.



วิทยากรรับเชิญ

คุณจรัส พันธุ์ทวี

ผู้พิพากษาศาลชั้นต้นประจำสำนักประธานศาลฎีกา



ดำเนินการเสวนาโดย

รศ.ดร.ชุตีสัตย์ เกิดวิบูลย์โส

ผู้อำนวยการสำนักเทคโนโลยีสารสนเทศ
สถาบันบัณฑิตพัฒนบริหารศาสตร์
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Thank you

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Thanks!



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